

National College of Ireland

Project Submission Sheet

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Programme: MSc in Artificial Intelligence **Year:** 2026
Module: Intelligent Agents and Process Automation
Lecturer: Prof. Shreyas Setlur Arun
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Project Title: Rental Application Processor: Automating the Rental Application Workflow Using RPA and AI-Driven Agents
Word Count: 1850 words

Academic Integrity Certificate Verification Code: ReRqtJ7dqn

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Signature: Harshwardhan Vijay Dongare
Date: 30/04/2026

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Intelligent Agents and Process Automation

CA

Your Name/Student Number	Course	Date
Harshwardhan Vijay Dongare	MSc in Artificial Intelligence	30/04/2026

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AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Consensus	Used for research and finding appropriate papers to reference and exploring methodologies used by other authors	https://consensus.app/
Perplexity	Used drawing a skeleton of report for latex overleaf	https://www.perplexity.ai/
Claude	To debug and to understand how to make an interface for better accessibility and understanding of project	https://claude.ai/

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

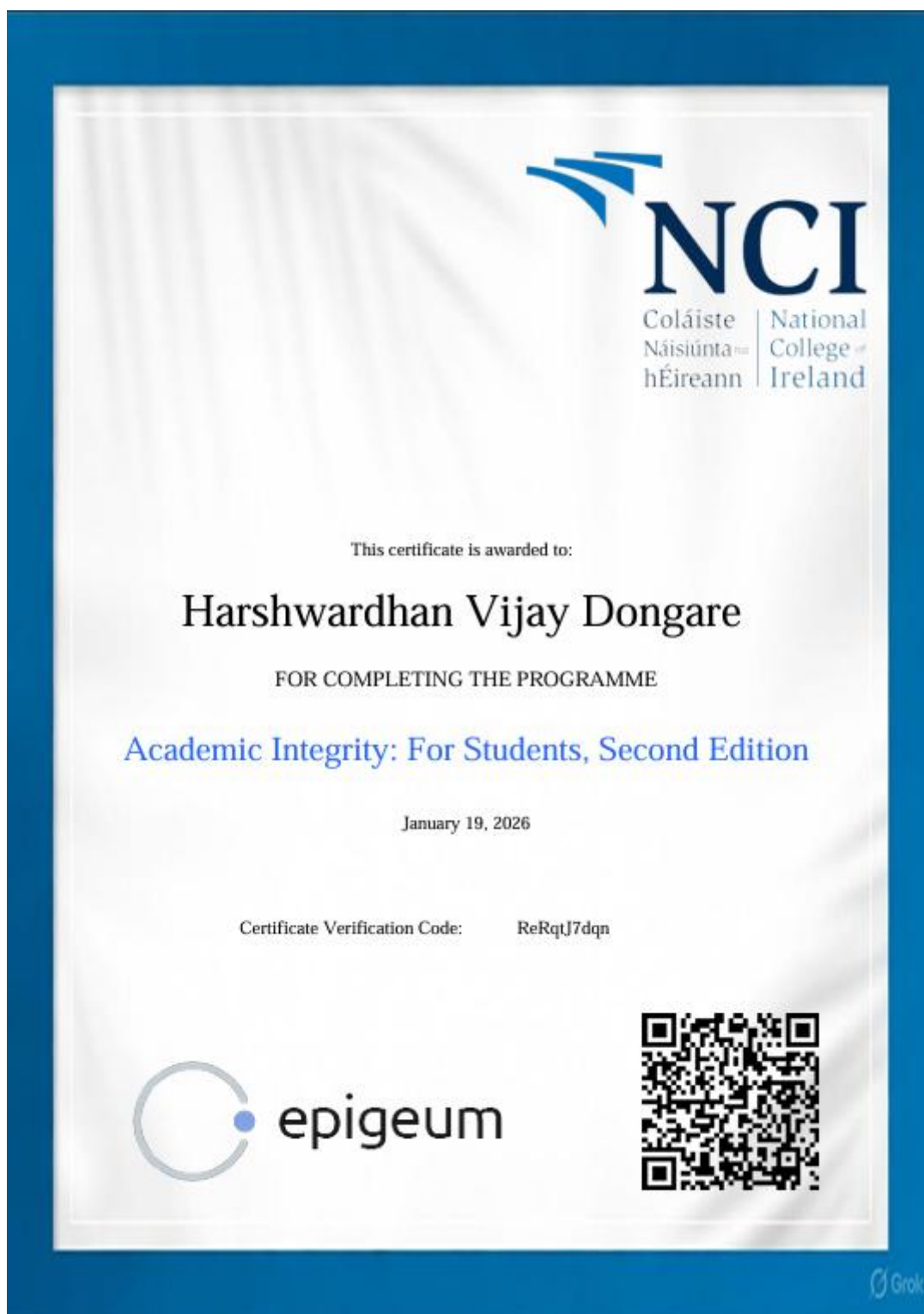
Perplexity	
Used for developing skeleton of latex code	
Can you give a outline skeleton in overleaf to report.	Here’s a clean Overleaf (LaTeX) report skeleton you can use for assignments, dissertations, or technical reports. It’s structured and ready to expand: <code>\documentclass[11pt,a4paper]{article}</code>
	 Quick Tips • Use Overleaf → New Project → Paste this code • Create a references.bib file for citations • Replace section names depending on your project (e.g., AI, data science, BPM, etc.)

Consensus	
Used for research and finding appropriate papers to reference and exploring methodologies used by other authors	
What are the current conditions of rental market in Dublin and are there any automations currently	Results 3M New <u>1</u> Housing Prices, Yields and Credit Conditions in Dublin since 1945 KEY TAKEAWAY·Sale prices in Dublin have risen by 8.4% per year since 1945, while market rents have increased by 6.3% per year, with four major housing market cycles since 1945, driven by credit conditions and user costs. 2020· 8citations ·Richard Keely et al. <i>The Journal of Real Estate Finance and Economics</i> <u>2</u> Experiencing housing precarity in the private rental sector during the covid-19 pandemic: the case of Ireland KEY TAKEAWAY·The Covid-19 pandemic has intensified housing precarity in Ireland's private rental sector, highlighting the need for better regulation, security, and quality in the sector. 2022· 44citations ·R. Waldron

Claude	
To debug and to understand how to make an interface for better accessibility and understanding of project	
I have error which says “ RecursionError: maximum recursion depth exceeded while calling a Python object ” what are the problems which might trigger this while im trying to use streamlit since im trying to integrate agents	Streamlit has several unique patterns that trigger this error beyond generic Python recursion. Given your project uses Streamlit with AI agents, here are the Streamlit-specific culprits: 1. st.session_state circular references Storing an object in session state that references itself or contains deeply nested structures from your AI agent responses. <pre>python # Dangerous — composite scorer result contains nested dicts referencing back st.session_state["result"] = composite_score_obj # if obj has circular refs</pre>

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Rental Application Processor

Automating the Rental Application Workflow Using RPA and AI-Driven Agents

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Course: MSc Artificial Intelligence **Module:** Intelligent Agents and Process Automation
National College of Ireland, April 2026

Abstract

This project report describes the Rental Application Processor, a hybrid automation system that handles the rental application process. The Rental Application Processor has two layers - a rule-based RPA layer for structured application intake, and an AI agent layer that uses a locally installed large language model (LLM) for qualitative document review. Integrated via a Streamlit web dashboard, these combine to transform 30 synthetic rental applications for five properties from documents to a shortlist, reducing the estimated 6-8 hours of workload for each property to 15-20 minutes of review time for an agent. The project also explores GDPR considerations, bias audits and how the EU AI Act classifies housing-related AI as a high-risk AI system.

1 Process Identification

The Dublin rental market is a tough one. An inner Dublin two-bed apartment will now rent for between 2,200 and 2,600 euro per month according to the Residential Tenancies Board Q4 2024 Rent Index [1]. An agent may receive up to 80-120 applications in less than 48 hours once a rental property is advertised. To read all of that, confirm the income levels, and try and recall who had the best references, would take an average agent six to eight hours per apartment to work through, simply in the initial round.

In this project, I opted to automate the rental application process. It begins with the applicant sending in their documents and finishes with the agent having a set of applicants in a ranked list to issue viewing invitations. I chose this workflow to work in because it's at the crossroads of two very different types of work. Determining whether a document is missing and calculating whether an applicant's salary is 3x their rent are both entirely rule-based. By reading a letter of recommendation for weasel words, or determining whether an applicant's cover letter demonstrates commitment takes more art. That division lends itself to a dual system which uses RPA and an AI agent for the quantitative attributes respectively [2].

This work resulted in the Rental Application Processor (Application Processor), a Python-based system which has two layers of automation, a six-screen Streamlit user interface for review and integrated GDPR and Fairness Auditing components. We tested the system using 30 realistic applications for five Dublin properties.

2 Process Modelling

2.1 As-Is Process: Manual Workflow

At present, most Dublin letting agencies have a completely manual and unstructured process [3]. The applicant emails a variety of documents in a variety of formats and the agent performs all of the tasks manually, without a defined rubric, and without a systematic approach.

The As-Is process is represented in a BPMN diagram in Figure 1. There are two swim lanes for the applicant and the agent. The process is completely sequential: the email comes, the agent manually checks if all documents are present, then calculates the income-to-rent ratio themselves, reads through the reference letters and then makes their own scoring decision. This process involves no automation. The only decision point in the process comes at the end when the agent decides which ones to shortlist based on their "gut feel". There's no time limit, no strict threshold and no record is kept. This essentially means that two agents presented with the exact same set of applications are likely to shortlist different people.

As-Is Process (Manual)



Figure 1: As-Is BPMP: manual two-lane process with single subjective decision gateway and no automated tasks and timers.

The bottlenecks in the process are apparent. The completeness of documents is not enforced, and therefore not always checked before moving to the next step. The income cap is interpreted differently by different agents or even overlooked when agents are busy. Referrals are skimmed fast when there is pressure, so warning signs in the wording, or too glowing or vague endorsements, may not be spotted. There is also no audit of why a particular applicant was selected or not, which presents a risk under the Equal Status Act 2000.

Cost Impacts

An agent that has five properties on his books spends 30-40 hours on on initial screening prior to viewing. At an average Dublin agent rate time of 20-25 euro an hour, the amount spent per property is 120-200 euro in wages time alone, not factoring in the time that is lost with clients that period [3].

Error Rates

Without checklists built into the system, checks for document completeness are between 15 to 20 percent of the high-volume cases. Income ratio calculations done by hand have the additional risk of calculation error; and even if agents reach the same result for a given case, interpretation of the thresholds may vary. varies between agents. Studies of manual, repetitive tasks show error rates of up to 10 percent when speedy responses are required [2]. In rental and rental assessment, a one-off mistake in progressing applicants time at every subsequent stage and may expose the property manager to a breach of the Equal Status Act 2000.

Lost Productivity

When the agent is searching for documents they are not negotiating property or landlord relationships. There also is no audit trail, so that if an unsuccessful applicant queries a decision, the agent cannot produce rationale to produce.

2.2 To-Be Process: Automated Pipeline

The new process shifts the swim lanes to four: applicant, an RPA bot layer, an AI agent layer and letting agent. Responsibility for actions is now spread across the lanes, as illustrated in Figure 2.

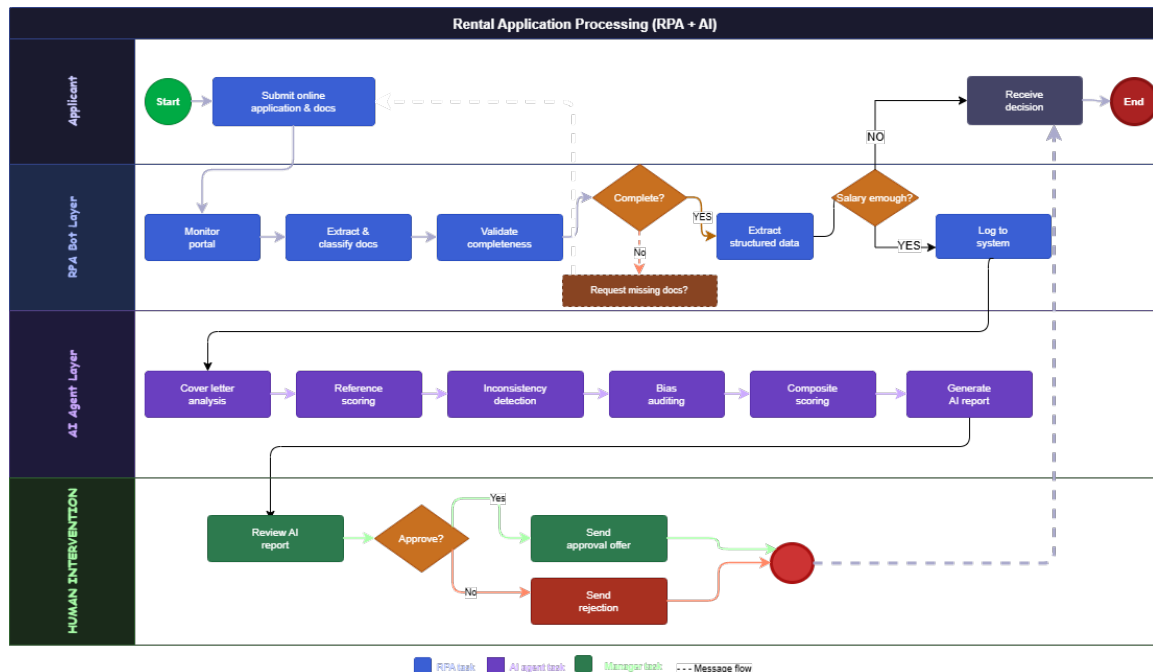


Figure 2: To-Be BPMN: four lanes, RPA intake and income gates, AI agent pipeline, and 48-hour timer event for uncomplete applications.

As soon as a new application is received, the RPA bot layer kicks in. It logs the application with a reference number and checks the six documents are all there, and if not, sends an email notification and a 48 hour timer event to resubmit. Successful applications are then automatically calculated for income. Any that are less than 2.75x rent cost will be automatically declined with a notification email that also advises the applicant that they can request a review by a real person. Those between 2.75x and 3.0x are noted for the agent's attention but still move on down the line. These are all passed as normalised JSON documentations for analysis by the AI layer.

Once this is done, the AI agent layer orchestrates a series of analyses. First, cover letter analysis, then employer reference analysis, landlord reference analysis, cross-document inconsistencies, composite score and finally summary report. All the letting agent sees is the result of all this analysis. They get a ready-ranked shortlist with summaries already prepared for each applicant and make the decision about who gets to meet for a viewing. That's a decision that remains with the human agent.

The shift from two lanes to four is significant not just structurally but also in terms of accountability. Each layer now has a clearly defined scope: the RPA layer makes to follow the rules

consistently in which the AI layer provides qualitative intelligence, and the human layer carries the legal weight of the final decision [4].

Cost Improvement

Our To-Be design targets each of the cost and error items listed above. On cost, the agent's in-house time is now reduced to around 15-20 minutes time in screening a shortlist per property. Under the same pay grade, this means the overall cost for screening per property to a modest 17 to 23, and recoups more than 80 percent of the cost the equivalent of 350 agent-hours per month. At a 50 property agency this gains in the order of 350 agent hours a month, which can be leveraged to increase revenue.

Error Reduction

On the errors front, the RPA layer addresses the two most frequent errors. The income test is now a computer test, so all of the required files have to be on until the six files are retrieved. The income threshold is applied identically to each applicant regardless of agent workload. The AI layer overcomes the skimming problem by reading all of the letters, and noting inconsistent information and hedging, lack of specificity and inconsistencies that a human reviewer with a likely to miss. The JSON result for each applicant can be checked, so any decision can be justified retrospectively.

Productivity Gains

With the intake and scoring process automated, the agent's time can now be ranking to ranking. The dashboard presents a ready-ranked list of recommendations for which the AI has crafted narratives, so the agent only at the point where human judgement plays a role.

Automation Recommendation

A strictly rule-based system would process intake quickly, but disregard soft cues in reference letters. A purely AI system would be un-auditable required for structured tasks. The hybrid system allocates each to the system most appropriate for it, keeps the human agent in the in charge of the final decisions and logs at each step [2].

3 Automation Potential Analysis

Table 1 evaluates each task in the redesigned process. Tasks 1 to 9 have deterministic, rule-based outcomes and are handled by the RPA layer. Tasks 10 to 15 require natural language understanding and are handled by the AI agent. Tasks 16 to 18 carry legal accountancy and remain with the human agent throughout.

Table 1: Automation potential analysis for tasks in the rental application workflow.

#	Task	Auto	Layer	Justification and Risk
1	Receive and register applications	Yes	RPA	Structured intake, deterministic, low risk
2	Assign reference numbers	Yes	RPA	Sequential numbering, no judgment required
3	Check document completeness	Yes	RPA	Binary file-presence check, low risk
4	Send acknowledgement emails	Yes	RPA	Templated and event-triggered, low risk
5	Request missing documents	Yes	RPA	Template with 48-hour timer event, low risk
6	Calculate income-to-rent ratio	Yes	RPA	Arithmetic rule applied to fixed threshold
7	Auto-reject below income threshold	Yes	RPA	Rule-based; all rejections reviewable by agent
8	Update master tracking spreadsheet	Yes	RPA	Structured data write, no ambiguity
9	Log GDPR consent to database	Yes	RPA	Timestamp-driven, fully deterministic
10	Score employer reference letter	Yes	AI	Tone, specificity, and red flag detection need NLP
11	Score landlord reference letter	Yes	AI	Tenancy history signals, medium risk
12	Analyse cover letter	Yes	AI	Motivation, stability, and sentiment analysis
13	Cross-document inconsistency check	Yes	AI	Pattern matching across five documents
14	Generate composite applicant score	Yes	AI	Weighted multi-factor aggregation
15	Post-hoc fairness audit	Yes	AI	Statistical analysis with LLM-generated narrative
16	Final viewing invitation decision	No	Human	Legal liability under Equal Status Act 2000
17	Property viewing coordination	No	Human	Real-time scheduling and agent relationship
18	Tenancy agreement sign-off	No	Human	Legal instrument, cannot be delegated

One design choice worth explaining is why Tasks 10 to 14 are handled by the AI layer rather than by simpler keyword-matching rules. A reference letter that never mentions reliability, or one that describes someone as having “improved significantly” over time, signals past problems that no keyword rule would catch reliably. A large language model reads those intentions and subtle framings naturally, because it has seen enough language to understand what a genuinely enthusiastic recommendation looks like versus a cautious one [5].

4 Automation Proposals

4.1 RPA-Based Automation

Tasks between 1 and 9 are encompassed in the RPA layer. Instead of an enterprise system such as UiPath or Automation Anywhere, pure Python was used with standard library modules, proving the same principles of RPA but without licensing complexities since it uses only pure Python and standard library modules in the implementation, featuring the same notion of RPA principles being practical and easy to understand (Vanderaalst 2018).

The whole intake-to-eligibility process is a sequence, orchestrated stream. Each ZIP archive uploaded via the dashboard is then unpacked by the system, a reference number deviating of REF-2026-XXX is assigned to each folder of an applicant and a tracker row, as well as a GDPR consent record with a 90-day deletion period, are written to a SQLite database. Each folder is then checked by a document validation pass which compares each one against a set of six files which are required: the application form, cover letter, employer reference, landlord reference, proof of income, and a photo ID flag. In case of the missing ones, an email notification will be designed and a 48 hour resubmission span will be provided.

After the completeness is verified, the income check reads the reported amount of gross monthly income offered in the application CSV, compares it with the amount in the payslip, and compares the ratio of the rent it covers. In 3.0x or more applications are sent to the AI layer. Anywhere between 2.75x and 3.0x are marked but sent anyway. Anyone with a rating below 2.75x is automatically rejected by email with the right to request human review. A master tracker in Excel is maintained on a regular basis and all emails that are sent are written to an audit log. A snippet of the terminal output of an entire pipeline execution appears in Figure 3, including document check execution results, and the output of the email simulation of each reference number.

```
[DOC CHECK] REF-2026-003 [OK] All documents present
[DOC CHECK] REF-2026-004 [OK] All documents present
[DOC CHECK] REF-2026-005 [OK] All documents present
[DOC CHECK] REF-2026-006 [OK] All documents present
[DOC CHECK] REF-2026-007 [OK] All documents present
[DOC CHECK] REF-2026-008 [FAIL] Missing: landlord_reference.txt
[EMAIL SIMULATION] Sent Missing Docs email to darragh.o'brien@example.com
[DOC CHECK] REF-2026-009 [OK] All documents present
[DOC CHECK] REF-2026-010 [OK] All documents present
[DOC CHECK] REF-2026-011 [OK] All documents present
[DOC CHECK] REF-2026-012 [OK] All documents present
[DOC CHECK] REF-2026-013 [OK] All documents present
[DOC CHECK] REF-2026-014 [OK] All documents present
[DOC CHECK] REF-2026-015 [OK] All documents present
[DOC CHECK] REF-2026-016 [OK] All documents present
[DOC CHECK] REF-2026-017 [OK] All documents present
[DOC CHECK] REF-2026-018 [OK] All documents present
[DOC CHECK] REF-2026-019 [OK] All documents present
[DOC CHECK] REF-2026-020 [OK] All documents present
[DOC CHECK] REF-2026-021 [OK] All documents present
[DOC CHECK] REF-2026-022 [OK] All documents present
[DOC CHECK] REF-2026-023 [OK] All documents present
[DOC CHECK] REF-2026-024 [OK] All documents present
[DOC CHECK] REF-2026-025 [OK] All documents present
[DOC CHECK] REF-2026-026 [FAIL] Missing: photo_id_present.txt
[EMAIL SIMULATION] Sent Missing Docs email to kerry.condon@example.com
[DOC CHECK] REF-2026-027 [OK] All documents present
[DOC CHECK] REF-2026-028 [OK] All documents present
```

Figure 3: Terminal output from a pipeline run showing document validation results (OK and FAIL), reference numbers, missing file names, and simulated email dispatch events.

The primary advantage of such a layer is uniformity: all applicants are evaluated based on precisely the same set of rules on the same basis. The primary practical issue is the rigid document schema demand and a production version of this system would require structured web intake form to implement it.

4.2 AI-Driven Automation

Tasks 10-15 are processed by an AI agent layer which uses a locally hosted LLaMA 3 model made available via an Ollama endpoint. Prompts are centralised into a single configuration file, and all calls emit structured JSON, which makes the outputs parsing outright by the dashboard, not requiring any intervention later.

The pipeline is implemented as a consecutive chain of call. Analysis of cover letter is first. A prompt is used to evaluate the motivational and stability cues and expressly does not evaluate the shorter and simpler letters of applicants whose first language may not be English. Then the employer and landlord references are individually scored. Questions to the model are to identify particular, verifiable praise and generic or hedging language, and each call is scored on 1 to 10 as well as reported red flags, and positive indicators. The inconsistency detector is then fed with all 5 document texts collectively and looks to identify any discrepancies in income, dates of employment as well as reasons why they moved according to the reason given and then categorizes the revelation as Low, Medium, or High risk.

After gathering the individual component scores, the composite scorer then adds all the scores together in a weighted formula:

$$S = 0.25S_{\text{inc}} + 0.20S_{\text{emp}} + 0.15S_{\text{land}} + 0.15S_{\text{cl}} + 0.15S_{\text{cons}} + 0.10S_{\text{docs}} \quad (1)$$

The income score is called S_{inc} . It is a number from 0 to 100. This score is based on how much rent's covered. There are also scores like S_{emp} and S_{land} . These scores are like a report card. They are based on a score from 1 to 10. The cover letter score is called S_{cl} . It is also based on a score from 1 to 10. The S_{cons} score starts at 100. If there is a risk it goes down to 70. If there is a risk it goes down to 40. There are five things to consider. The S_{docs} score starts at 100. Goes down by 10 points for every document that is missing. The model is used again to make a summary that is 150 words long for the agent. This system is good because it can find things that might be missed when people are, in a hurry. It also makes it easy to see where all the scores come from because it makes a JSON file that's easy to understand. The main problem is that the scores can be different each time the model is used. To make this problem smaller the system checks the JSON file to make sure it is correct before it accepts any scores. It also makes sure that every prompt is based on the text of the document not just the instructions.

5 Solution Implementation

5.1 System Architecture

The complete Application Processor architecture is indicated in 4. The system has three horizontal levels, where data flows downward, starting with the input to the processing to the outputs.

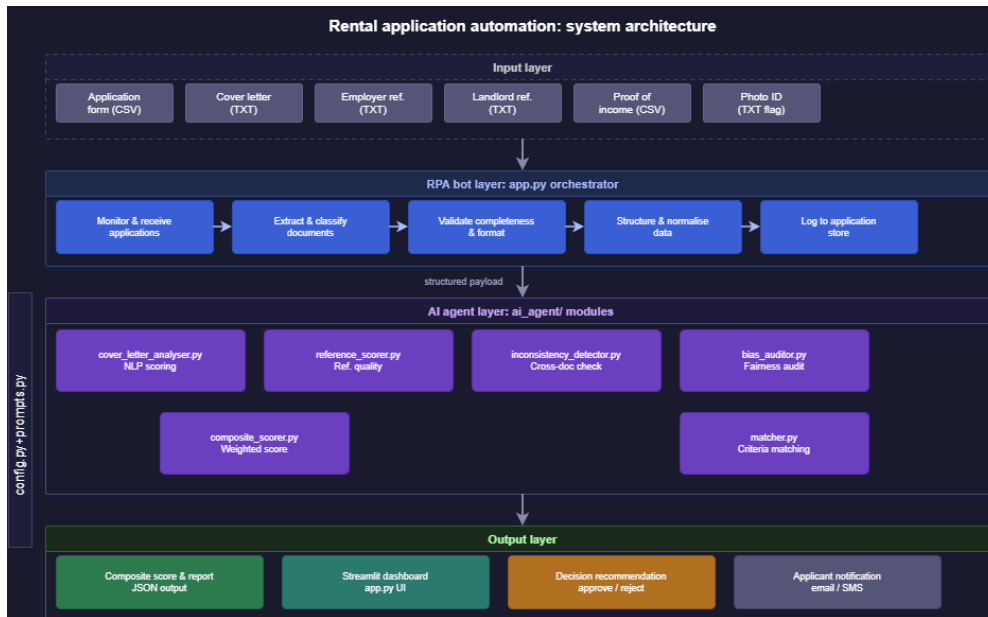


Figure 4: application processor system architecture: input layer, which receives six types of documents, RPA bot layer, consisting of five consecutive steps, AI agent layer, comprising of six analysis modules, and four output delivery structures.

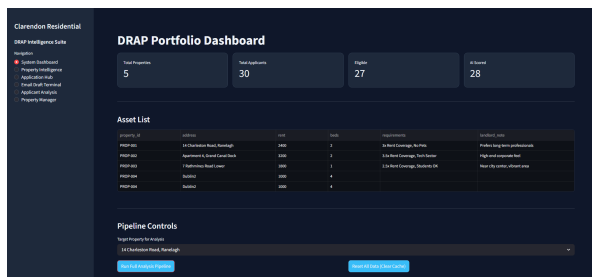
There are six types of documents that are accepted in the input layer (one applicant per document type): an application form (csv format), a cover letter (plain text), an employer reference letter (plain text), a landlord reference letter (plain text), a proof of income (csv format), and a photo ID presence flag. The RPA bot layer concludes therein, and is an orchestrated process: the bot monitors new submissions, retrieves and categorises documents, verifies completeness and formatting, formats and normalises the data and documents all of that in the master tracker. The layered output of this layer is the structured payload, which is then fed into the layer of AI agents executing six modules, to generate component scores, composite score, and narrative summaries. The output layer returns the results in a variety of forms; the individual JSON report cards, ranked shortlist CSV, a dashboard decision recommendation screen, and email notifications to the applicant.

The entire pipeline is launched on the command line. RPA phases perform sequentially first and all the qualified applications are sent through AI layer as a batch. A post-hoc audit is a stand-alone command. This was initially provided because the EU AI Act categorizes AI in housing at the entry as high risks, in terms of the use in the Annex III, and patterns of discrimination should be monitored continuously in this area of application. The system uses Python 3.11. Some other tools like Streamlit 1.35 pandas and openpyxl. It also uses the OpenAI- SDK to talk to a local Ollama LLaMA 3 instance. For storing information the system uses SQLite for the GDPR ledger.

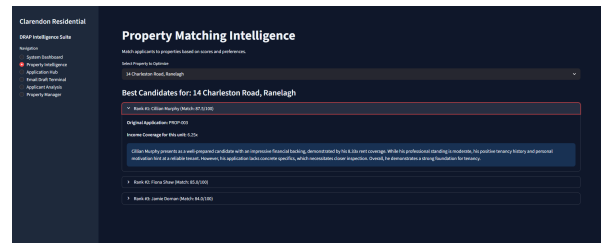
All the settings like score weights, income thresholds and email mode are, in one file. This makes it easy to change the system for any property without having to change the code.

5.2 Streamlit Web Interface

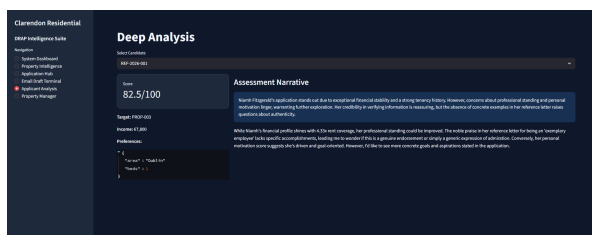
The dashboard gives the letting agent a six-screen interface to review results without having to open any JSON files or spreadsheets directly.



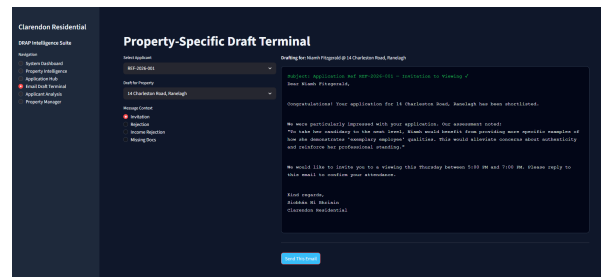
(a) System Dashboard: portfolio summary showing 5 properties, 30 applicants, 27 eligible, 28 AI-scored; asset list and pipeline trigger controls.



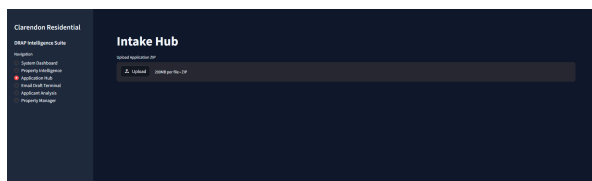
(b) Property Matching Intelligence: ranked candidates per property. Top match for 14 Charleston Road scores 87.5/100 with 6.25x income coverage.



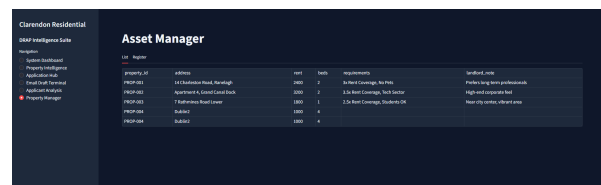
(c) Applicant Deep Analysis: REF-2026-001 scores 82.5/100 on a monthly income of 7,800. Narrative flags a strong financial profile but questions the specificity of the employer reference.



(d) Email Draft Terminal: AI-generated contextualized emails for invitations, rejections, income rejections, and missing document requests.



(e) Intake Hub: ZIP upload interface. The RPA layer extracts, validates, and registers all folders automatically, supporting uploads up to 200 MB.



(f) Asset Manager: five-property register with rent, bed count, landlord requirements and notes. A Register tab adds new properties to the portfolio.

Figure 5: All six screens of the Application Processor Intelligence Suite Streamlit dashboard.

The System Dashboard (Figure 5a) provides an overview of the portfolio with pipeline triggers. The Property Matching Intelligence (Figure 5b) tab displays the top matches for each listing with the income covered. The Applicant Deep Analysis screen (Figure 5c) shows a breakdown of the scores and AI narrative for a single applicant. The Email Draft Terminal (Figure 5d) tools offer context-sensitive emails for four email types, and they're automatically pre-filled with the applicant name, address of the property, and any relevant information from the AI summary. The Intake Hub (Figure 5e) allows ZIP uploads. The Asset Manager (Figure 5f), manages the property register and the agent can add new properties to be managed without needing to change code.

6 Ethical Considerations

Housing is one of the most obvious areas where the decisions made by computers have a direct impact on individuals. It's not an abstract concept of harm to be denied housing on the basis

of an unknown and unexplainable algorithmic calculation. This informed our design decisions about Application Processor.

The General Data Protection Regulation (GDPR) Article 22 gives the right not to be subject to solely automated decisions that have a significant impact on the individual. Application Processor uses the composite score and ranking as suggestions. The agent always makes the final invitation decision, and all rejected applications are sent an email stating the right to request review [6]. In compliance with GDPR Article 13, the acknowledgement email states that automated processing is in operation. Consent records are programmed into a SQLite ledger to be deleted 90 days after tenancy decisions.

The score audit module tests for scoring patterns that could be used as a proxy for protected features under the Irish Equal Status Act 2000. For example, it tests for differences in score distributions between perceived writing styles and letter complexity, which may serve as a proxy for nationality. This also explicitly includes a prompt in the cover letter about not rewarding shorter, simpler writing from people for whom English may be a second language. This is an explicit, but far from perfect, effort to minimise the risk of proxy discrimination at this stage.

The EU AI Act assigns the use of AI in housing to the high-risk category according to Annex III, which calls for continuous checks for discrimination patterns and the documenting of the transparency notices to applicants [7]. The ex-post audit and human gate to decision and the acknowledgement email transparency statement are the minimum compliance structure required for this classification. In a real deployment more audit logs and documentation would be required, but the compliance framework is in place.

7 Conclusion and Limitation

Application Processor demonstrates how an RPA and AI system can help the rental applications process. The RPA component addresses the rule-based component at scale and speed that would be unviable to do it manually with 100 applications in 48 hours. The AI layer provides the linguistic skills to deal with the soft documents' language and picks up reference issues and inconsistencies that might otherwise be overlooked in a quick human scan. Together they reduce the agent's time for a given property from 6 to 8 hours to 15 to 20 minutes reviewing a short-list.

It's all written in Python, tested on 30 synthetic applications for a property sample of five Dublin properties, and the user experience is a six-screen Streamlit application. Privacy (consent, logging to GDPR requirements, transparency disclosures for the applicant), and a post-doc review for biases are all integral to the design. All scores are transparent in the structured JSON output, and the agent can always report how a decision was made.

Of course, there are limitations such as the need for structured documents, which would require a structured data web form in production applications, and the non-deterministic nature of the LLM processing, which is somewhat constrained by the requirement for validated output values. Future enhancements might apply the system to unstructured email submissions through the addition of a Natural Language Processing document classifier, and replace the on-premise llama 3 model with an Irish-specific rental application language model that it has been fine tuned to.

8 Files to be submitted

All the files have be stored in the drive: [CLICK ON TEXT TO REDIRECT](#)

References

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